

Subject: Is it OK for AI to write science papers?
Date: Wednesday, May 14, 2025 at 10:56:47 AM Pacific Daylight Time
From: Pinheiro-Chagas, Pedro
To: MAC - AI
BCC: Sirota, Marina
Attachments: ai_scientific_writer_methods.pdf

Hi MAC AI list +,

I recently started developing an agentic AI system (with humans in the loop) – AI Scientific Writer (https://github.com/pinheirochagas/AI_scientific_writer) - to convert scientific talks into publication-ready review articles, see attachment. It can process audio recordings and/or presentation slides. We have already submitted the first paper *Redefining Language and Neurodegeneration Through PPA: Clinical Phenotypes, Network Vulnerability, and Global Research Directions* with Yu-Wen Cheng, Maria Luisa Gorno-Tempini and Boon Lead Tee. I will share it here once published! The next step would be to expand this framework to make review papers live, by automating a system to periodically check for relevant new published manuscripts and integrate them into the main text.

While I was working on this project, I began thinking quite a bit about the ethics of AI writing papers. For me, the three crucial needs are: humans in the loop to verify the generated text and suggested references, fully disclosing how exactly AI was used in the manuscript and disclosing the prompts used. All these features are included in the AI Scientific Writer tool.

The most extreme version of tools like this is the AI Scientist-v2 from Sakana Labs (<https://sakana.ai/ai-scientist-first-publication/>), which is a fully automated end-to-end AI system autonomously generates hypotheses, runs experiments, analyzes data, and writes scientific manuscripts (across Machine Learning ML domains). Their paper achieved high review scores at ICLR (one of the 3 major ML conferences). Read their technical report here (<https://arxiv.org/abs/2504.08066>) and check the open-sourced code here (<https://github.com/SakanaAI/AI-Scientist-v2>).

See below some very interesting perspectives fresh from Nature, as well as their AI Research test – take it!

Poll of 5,000 researchers finds contrasting views on when it's acceptable to involve AI and what needs to be disclosed.

<https://www.nature.com/articles/d41586-025-01463-8>

What's your view on using AI for peer review and for writing research papers?

<https://www.nature.com/immersive/d41586-025-01512-2/index.html>

Another interesting perspective:

<https://www.nature.com/articles/d41586-025-00894-7>

SHARE YOUR THOUGHTS in this document!

https://docs.google.com/document/d/1WevfX8DxSwvADnNfOxiMfR35T_hKf6Xw73i0eOiDPEs/edit?tab=t.0

Best,

Pedro

--

[Pedro Pinheiro-Chagas, PhD](#)

Assistant Professor – [NeuroAI Lab](#)

University of California, San Francisco

Department of Neurology, [Memory and Aging Center](#)

UCSF Nancy Friend Pritzker Psychiatry Building

675 18th St, Office 3322 | San Francisco, CA 94107, USA

[Dyslexia Center](#) | [Multitudes](#) | [Math Cognition](#)

[Bakar Computational Health Sciences Institute](#)

[Center for Intelligent Imaging \(ci2\)](#)

[Weill Institute for Neurosciences](#)